Chapter 9. Discriminative Methods: Logistic Regression

Lecture 7-2 (discriminative vs generative, logit model, GLM, maximum likelihood, confounding, extensions, multinomial regression)

Contents

1 Discriminative Methods versus Generative Models **2**

2 Why Not Linear Regression? **2**

3 The Logistic Model **2**

3.1 Logistic regression as a GLM 2

4 Estimation by Maximum Likelihood **3**

5 Logistic Regression on the Default Data **3**

6 Multiple Logistic Regression and Confounding **3**

7 Extensions of Logistic Regression **4**

8 More Than Two Classes: Multinomial Regression **4**

9 Selected exercises **5**

1 Discriminative Methods versus Generative Models

Discriminative: model $\Pr(Y = c_k \mid X = x)$ directly	Generative: model (X, Y), then derive $\Pr(Y = c_k \mid X = x)$
K-nearest neighbors, logistic regression, classification trees, ensembles (bagging, random forest, boosting), support vector machines	linear and quadratic discriminant analysis, naive Bayes, Gaussian mixture models

Examples: tumours benign/malignant; transactions legitimate/fraudulent; news topics. Often we estimate $P(Y = k \mid X)$ and classify to the most probable class.

2 Why Not Linear Regression?

With $Y \in \{0, 1\}$ and $\hat{Y} = \beta_0 + \beta X$: $\mathbb{E}[Y \mid X] = P(Y = 1 \mid X) \in [0, 1]$, but a linear fit is unbounded. (For two classes the classification rule equals LDA's, but the probabilities are poor.)

Coding $Y = 1$ (stroke), 2 (drug overdose), 3 (epileptic seizure) imposes a meaningless ordering and spacing; different codings give different fits.

3 The Logistic Model

For $Y \in \{c_1, c_2\}$, the **logit** model:

$$\log \frac{\Pr(Y = c_1 \mid X)}{\Pr(Y = c_2 \mid X)} = \beta_0 + \beta_1 X,$$

$$\Pr(Y = c_1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}, \quad \Pr(Y = c_2 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}},$$

an S-shaped curve in $[0, 1]$. β_1: change in the **log odds** per unit of X; e^{β_1} multiplies the odds; $\partial p / \partial X = \beta_1 p(1 - p)$.

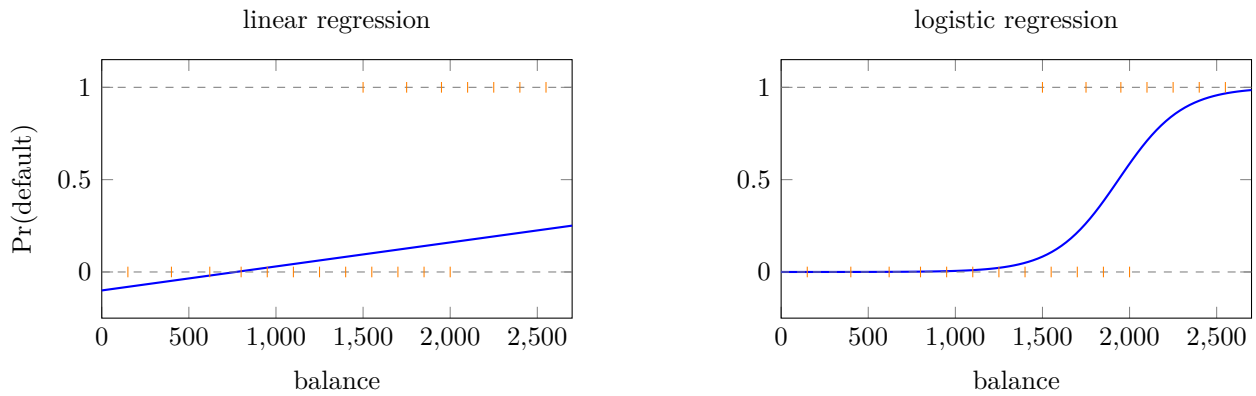


Figure 1: Default data (schematic). The linear fit leaves $[0, 1]$; the logistic fit $\hat{p} = e^{-10.65 + 0.0055X} / (1 + e^{-10.65 + 0.0055X})$ does not.

3.1 Logistic regression as a GLM

A GLM: Y from an exponential family, linear predictor $\eta = X^\top \beta$, link $g(\mathbb{E}[Y]) = \eta$. Here $Y \sim$ Bernoulli(p) with the canonical logit link, the natural parameter of $p^y (1 - p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1 - p)\}$.

4 Estimation by Maximum Likelihood

Motivation. No closed form: maximize the likelihood; standard errors come from its curvature.

$$\mathcal{L}(\beta_0,\beta)=\prod_{i:y_i=1}p(x_i)\prod_{j:y_j=0}\{1-p(x_j)\},\qquad \ell(\beta_0,\beta_1)=\sum_i\Big[y_i(\beta_0+\beta_1x_i)-\log\left(1+e^{\beta_0+\beta_1x_i}\right)\Big].$$

Theorem: score equations and concavity

With $p_i=p(x_i)$ and design rows $\tilde{x}_i=(1,x_i)^\top$,

$$\nabla\ell=\sum_i(y_i-p_i)\tilde{x}_i=\tilde{X}^\top(y-p),\qquad \nabla^2\ell=-\tilde{X}^\top W\tilde{X},\ W=\mathrm{diag}\{p_i(1-p_i)\}.$$

ℓ is concave, so the MLE solves $\tilde{X}^\top(y-p)=0$ and is found by Newton–Raphson, equivalently **iteratively reweighted least squares**: $\beta^{\text{new}}=(\tilde{X}^\top W\tilde{X})^{-1}\tilde{X}^\top Wz$ with working response $z=\tilde{X}\beta+W^{-1}(y-p)$.

Proof [Supplement]. $\frac{d}{d\eta}\log(1+e^\eta)=\frac{e^\eta}{1+e^\eta}=p$ and $\frac{dp}{d\eta}=p(1-p)$; differentiate termwise. The Hessian is negative semidefinite. A Newton step $\beta+(\tilde{X}^\top W\tilde{X})^{-1}\tilde{X}^\top(y-p)$ rearranges into the weighted least-squares form. ■

$\widehat{\mathrm{Var}}(\hat{\beta})=(\tilde{X}^\top\hat{W}\tilde{X})^{-1}$ (inverse Fisher information) gives SEs and z-statistics. Under perfect separation the MLE does not exist (Exercise 9.1; Ch. 10).

5 Logistic Regression on the Default Data

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.6513	0.3612	−29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

One unit of balance adds 0.0055 to the log odds; $H_0:\beta_1=0$ is rejected ($p<0.0001$).

Predictions.

$$\hat{p}(1000)=\frac{e^{-10.6513+0.0055\times1000}}{1+e^{-10.6513+0.0055\times1000}}=0.006,\qquad \hat{p}(2000)=\frac{e^{-10.6513+0.0055\times2000}}{1+e^{-10.6513+0.0055\times2000}}=0.586.$$

A qualitative predictor. With student status: intercept −3.5041, student[Yes] 0.4049 ($p=0.0004$), so $\Pr(\text{default}\mid\text{student})=e^{-3.0992}/(1+e^{-3.0992})=0.0431$ and $\Pr(\text{default}\mid\text{non-student})=e^{-3.5041}/(1+e^{-3.5041})=0.0292$.

6 Multiple Logistic Regression and Confounding

$$\log\frac{p(X)}{1-p(X)}=\beta_0+\beta_1X_1+\cdots+\beta_pX_p,\qquad p(X)=\frac{e^{\beta_0+\beta_1X_1+\cdots+\beta_pX_p}}{1+e^{\beta_0+\beta_1X_1+\cdots+\beta_pX_p}}.$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	−10.8690	0.4923	−22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income (thousands)	0.0030	0.0082	0.37	0.7115
student[Yes]	−0.6468	0.2362	−2.74	0.0062

The student coefficient is now *negative*. **Confounding**: students carry higher balances, so their marginal default rate is higher, but at fixed balance they default *less*.

7 Extensions of Logistic Regression

- **Polynomial logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d$. E.g. $\texttt{wage} > 250$ on a degree-4 polynomial in age.
- **Piecewise-constant logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i)$ with step functions of age.
- **GAM for classification**: $\log \frac{P(X)}{1-P(X)} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p)$; for the Wage data, $\beta_1\,\texttt{year} + f_2(\texttt{age}) + f_3(\texttt{education})$ with a smoothing spline (5 df) in age and a step function for education.
- **Penalized logistic regression**: minimize $-\log \mathcal{L}(\beta_0, \beta) + \lambda\|\beta\|_2^2$ (ridge) or $+\lambda\|\beta\|_1$ (lasso), generally $+\lambda\|\beta\|_q^q$; the penalty also gives finite estimates under separation.
- **Multinomial logistic regression** (below).

8 More Than Two Classes: Multinomial Regression

Baseline class K; for $k = 1, \ldots, K-1$,

$$\log \frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} = \beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p.$$

Theorem: implied probabilities

$$\Pr(Y = k \mid X = x) = \frac{e^{\beta_{k0} + \beta_k^\top x}}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_l^\top x}} \quad (k < K),$$

$$\Pr(Y = K \mid X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} e^{\beta_{l0} + \beta_l^\top x}},$$

and the log odds between any two classes k, k' is linear: $\log \frac{\Pr(Y=k|x)}{\Pr(Y=k'|x)} = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$.

Proof [Supplement]. Write $\pi_k = \Pr(Y = k \mid x) = \pi_K e^{\eta_k}$ for $k < K$; the probabilities sum to 1, so $\pi_K(1 + \sum_l e^{\eta_l}) = 1$. The difference of two log odds against K gives the pairwise log odds. ■

The baseline affects only the parametrization. Softmax: $\Pr(Y = k \mid x) = e^{\eta_k} / \sum_{l=1}^{K} e^{\eta_l}$, identifiable up to a common shift (Exercise 9.4).

Summary

- Logistic regression: linear log odds (Bernoulli GLM, logit link), fitted by IRLS.
- Multiple regression reveals confounding; multinomial regression has linear pairwise log odds.

9 Selected exercises

Motivation. Separation, a binary predictor, and ISLR Ex. 4.6 and 4.12.

Exercise 9.1 (No MLE under complete separation)

Let $\ell(\beta) = \sum_i [y_i \tilde{x}_i^\top \beta - \log(1 + e^{\tilde{x}_i^\top \beta})]$ with $y_i \in \{0, 1\}$, and suppose some b satisfies $\tilde{x}_i^\top b > 0$ whenever $y_i = 1$ and $\tilde{x}_i^\top b < 0$ whenever $y_i = 0$.
(a) Show that $\ell(\beta) < 0$ for every β and $\ell(tb) \to 0$ as $t \to \infty$; hence no maximizer exists.
(b) Show that for $\lambda > 0$ the ridge-penalized objective $\ell(\beta) - \lambda\|\beta\|^2$ has a unique maximizer.

Solution [Supplement].

(a) With $p_i = e^{\tilde{x}_i^\top \beta}/(1+e^{\tilde{x}_i^\top \beta}) \in (0, 1)$, each term of ℓ equals $\log p_i$ if $y_i = 1$ and $\log(1-p_i)$ if $y_i = 0$, the log of a number in $(0, 1)$; so $\ell(\beta) < 0$. Along $\beta = tb$: if $y_i = 1$, $\tilde{x}_i^\top b > 0$ and $p_i \to 1$; if $y_i = 0$, $\tilde{x}_i^\top b < 0$ and $p_i \to 0$. Every term tends to 0, so $\sup_\beta \ell = 0$, which is never attained.

(b) $F(\beta) = \ell(\beta) - \lambda\|\beta\|^2$ is continuous, $F \le -\lambda\|\beta\|^2 \to -\infty$ as $\|\beta\| \to \infty$, so the set $\{F \ge F(0)\}$ is closed and bounded and F attains its maximum there. ℓ is concave (its Hessian is $-\tilde{X}^\top W \tilde{X}$) and $-\lambda\|\beta\|^2$ is strictly concave, so F is strictly concave and the maximizer is unique. ■

Exercise 9.2 (Logistic regression on one binary predictor)

Let $x_i \in \{0, 1\}$; group x has n_x observations of which s_x have $y_i = 1$, with $0 < s_x < n_x$, and $\hat{p}_x = s_x/n_x$. Fit $\log \frac{p}{1-p} = \beta_0 + \beta_1 x$.
(a) Show that $\hat{\beta}_0 = \log \frac{\hat{p}_0}{1-\hat{p}_0}$ and $\hat{\beta}_1 = \log \frac{\hat{p}_1/(1-\hat{p}_1)}{\hat{p}_0/(1-\hat{p}_0)}$ (the log odds ratio), so the fitted probabilities are the group proportions.
(b) Show that whenever the model has an intercept, $\sum_i \hat{p}_i = \sum_i y_i$.

Solution [Supplement].

(a)

Step 1. The score equations $\tilde{X}^\top (y - p) = 0$ read $\sum_i (y_i - p_i) = 0$ and $\sum_i x_i (y_i - p_i) = 0$.

Step 2. In group x all p_i equal $p(x) = \sigma(\beta_0 + \beta_1 x)$, $\sigma(\eta) = e^\eta/(1 + e^\eta)$. The second equation gives $s_1 - n_1 p(1) = 0$; subtracting it from the first gives $s_0 - n_0 p(0) = 0$. So $p(0) = \hat{p}_0$, $p(1) = \hat{p}_1$.

Step 3. Inverting σ yields $\hat{\beta}_0 = \text{logit}(\hat{p}_0)$ and $\hat{\beta}_0 + \hat{\beta}_1 = \text{logit}(\hat{p}_1)$. Both groups are present, so $\tilde{X}$ has full column rank and $W \succ 0$; the Hessian $-\tilde{X}^\top W \tilde{X}$ is negative definite, ℓ is strictly concave, and this solution is the unique MLE.

(b) The intercept's score equation is $\sum_i (y_i - \hat{p}_i) = 0$.

Default data: 0.0292 and 0.0431 are the non-student and student default rates; $e^{0.4049} \approx 1.50$ is their odds ratio. ■

Exercise 9.3 (Reading a logistic fit; ISLR Ex. 4.6)

For students, X_1 = hours studied, X_2 = undergraduate GPA and Y = receiving an A, a logistic fit gives $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.
(a) Estimate the probability of an A for a student who studies 40 hours and has GPA 3.5.
(b) How many hours would this student need for a 50% chance?

Solution.

(a) The linear predictor is $-6+0.05(40)+3.5 = -0.5$, so $\hat{p} = e^{-0.5}/(1+e^{-0.5}) = 1/(1+e^{0.5}) \approx 0.378$.

(b) $\hat{p} = 0.5$ iff the log odds are 0: $-6 + 0.05h + 3.5 = 0$, i.e. $h = 50$ hours. ■

Exercise 9.4 (Logistic versus softmax coding; ISLR Ex. 4.12)

To classify $X \in \mathbb{R}$ into apples and oranges, you fit $\Pr(\text{orange} \mid x) = e^{\beta_0+\beta_1 x}/(1 + e^{\beta_0+\beta_1 x})$; a friend fits the softmax model $\Pr(\text{orange} \mid x) = e^{\alpha_{o0}+\alpha_{o1}x}/(e^{\alpha_{o0}+\alpha_{o1}x} + e^{\alpha_{a0}+\alpha_{a1}x})$, both by maximum likelihood on the same data.
(a) Give the log odds of orange versus apple in both models.
(b) If $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = -1$, what are the friend's estimates?
(c) If the friend gets $\hat{\alpha}_{o0} = 1.2$, $\hat{\alpha}_{o1} = -2$, $\hat{\alpha}_{a0} = 3$, $\hat{\alpha}_{a1} = 0.6$, what are yours?
(d) How often will the two predicted labels agree on 2000 test points?

Solution.

(a) Yours: $\beta_0 + \beta_1 x$. The friend's: $\log \frac{e^{\alpha_{o0}+\alpha_{o1}x}}{e^{\alpha_{a0}+\alpha_{a1}x}} = (\alpha_{o0} - \alpha_{a0}) + (\alpha_{o1} - \alpha_{a1})x$.

Key fact. By (a), the softmax probabilities equal the logistic ones with $\beta = \alpha_o - \alpha_a$, so the friend's likelihood is $L_{\text{soft}}(\alpha) = L_{\log}(\alpha_o - \alpha_a)$. As α ranges over $\mathbb{R}^4$, $\alpha_o - \alpha_a$ ranges over all of $\mathbb{R}^2$; hence $\hat{\alpha}$ maximizes L_{soft} iff $\hat{\alpha}_o - \hat{\alpha}_a$ maximizes $L_{\log}$.

(b) Only differences are determined: $\hat{\alpha}_{o0} - \hat{\alpha}_{a0} = 2$ and $\hat{\alpha}_{o1} - \hat{\alpha}_{a1} = -1$, e.g. $\hat{\alpha}_a = (0, 0)$, $\hat{\alpha}_o = (2, -1)$, or any common shift of both.

(c) $\hat{\beta}_0 = 1.2 - 3 = -1.8$ and $\hat{\beta}_1 = -2 - 0.6 = -2.6$.

(d) Always (100%): the two fits give identical probabilities at every x, hence identical labels. ■